

# README

## BOB-096 — Chaos coverage for qBitTorrent-go/tests/ integration/jackett\_db\_test.go

**Ticket:** BOB-096 **Scope:** extend the boba-jackett DB integration suite with §11.4.85-class process-death, resource-exhaustion, and concurrent-kill chaos scenarios. **Date:** 2026-08-19 **Race detector:** clean (go test -race, 3× deterministic re-run).

### Scenarios added

| # | Scenario                                                                                  | Test function                          | §11.4.85 class |
|---|-------------------------------------------------------------------------------------------|----------------------------------------|----------------|
| 1 | Resource-exhaustion — file-descriptor pressure via os.Pipe() (bounded per §12.6)          | TestChaos_FileDescriptorExhaustion     | (d)            |
| 2 | Concurrent-kill via context.CancelFunc from a second goroutine while the first is writing | TestChaos_ConcurrentContextCancelWrite | (a-extra)      |

### Scenarios already present (kept, referenced for completeness)

| # | Scenario | Test function                           | §11.4.85 class |
|---|----------|-----------------------------------------|----------------|
| 3 |          | TestChaos_MidTransactionSIGKILLRecovery | (a)            |

| # | Scenario                                                        | Test function                            | §11.4.85 class |
|---|-----------------------------------------------------------------|------------------------------------------|----------------|
|   | Process-kill (SIGKILL of a helper subprocess mid-transaction)   |                                          |                |
| 4 | Byte-level DB file corruption                                   | TestChaos_DBFileByteCorruption           | (b)            |
| 5 | Concurrent-writer contention (40 goroutines, direct repo layer) | TestChaos_ConcurrentWriterContentionRepo | (c)            |
| 6 | WAL sidecar corruption + reopen                                 | TestChaos_WALSidecarCorruptionRecovery   | (a-analogue)   |
| 7 | Master-key rotation mid-flight (K1→K2 forward + K1 reverse)     | TestChaos_MasterKeyRotationMidflight     | (e-analogue)   |

## §11.4.115 RED-first evidence — captured live

For `TestChaos_ConcurrentContextCancelWrite`, invariant (ii) (“every confirmed write round-trips”) was proven falsifiable by injecting a phantom name `KILL_PHANTOM` into `finalNames` with no corresponding DB row.

- `RED_phantom_mutation.log` — captures the FAIL: (ii) confirmed-write `KILL_PHANTOM` lost after cancel: repos: credential not found → FAIL: `TestChaos_ConcurrentContextCancelWrite`
- Mutation reverted; `GREEN_after_restore.log` captures the immediate PASS under identical race conditions.

Falsification narratives for the other invariants (i, iii on FD + cancel tests; all invariants on the pre-existing tests) are recorded in the per-test doc-comments per §1.1.

## Anti-bluff assertions (§11.4.5 / §11.4.107)

- Byte-equal plaintext round-trip via `Credential.Username/Password`, not “no error”.
- `sql.DB` row count vs writer-confirmed count exact equality.
- Categorised error on FD-pressure Upsert failure (non-empty `err.Error()`).
- Durability boundary: BASELINE / PRECOMMIT rows always survive; uncommitted rows never surface.
- Panic guard: every DB access under chaos wrapped in `recover() → t.Fatalf`.

## Cleanup (§11.4.14)

- `TestChaos_FileDescriptorExhaustion` closes every `os.Pipe()` pair via both `defer cleanup()` and `t.Cleanup(cleanup)` (belt-and-braces).
- `TestChaos_ConcurrentContextCancelWrite` registers `t.Cleanup(cancel)` so the context is cancelled even on early test bailout.
- `t.TempDir()` handles DB + evidence-log cleanup automatically.

## Host safety (§12.6, §12.12, §11.4.161)

- FD pressure capped at `min(RLIMIT_NOFILE.soft / 4, 512)` — never destabilises the host FD table.
- No sudo, no rootful container, no host power-management calls.
- `GOMAXPROCS=2 nice -n 19 ionice -c 3` throughout.

## Determinism (§11.4.50)

Three consecutive `-race -count=3` runs — all GREEN (`race_detector_3x.log`).

## Run log files

- `bob096_run_verbose.log` — verbose PASS output for the two new tests.
- `full_chaos_suite.log` — full `TestChaos*` suite under `-race`.
- `new_chaos_scenarios.log` — first run of the two additions.
- `RED_phantom_mutation.log` — captured RED (invariant (ii) fails).
- `GREEN_after_restore.log` — GREEN immediately after restoring source.

- `race_detector_3x.log` — 3× deterministic race-clean run.

## **Skipped chaos class (honest, tracked)**

§11.4.85 (d) `disk-full` remains deferred as a follow-up §11.4.197 item — a `filesystem-quota` fixture belongs in a container-scoped harness rather than a host-shared test file. Recorded in the test-file doc-comment.